Chapter 93 — Large List Performance

> **Part X — Performance, Observability & Quality** · Complexity: **Standard** (≥25 interview questions)
>
> **Prerequisites:** Ch 90 (jank/recomposition), Ch 71 (infinite feed/Paging 3), Ch 59 (image loading), Ch 89 (memory), Ch 85 (perf monitoring). Pairs with Ch 88 (cold start/first scroll), Ch 92 (baseline profiles), Ch 2 (Compose).
>
> **Reference surface in StreamCart:** scrolling **large lists/grids** (product catalog, feed Ch 71, order history) **smoothly** — via `LazyColumn` / `LazyRow` / `LazyVerticalGrid` (view recycling) with **stable** `key`**s**, `contentType`, **light/stable item composables** (Ch 90), **async image loading** (Coil, Ch 59), and **paging** (Ch 71) — so even thousands of items scroll within the **frame budget** (Ch 90) without jank, excess memory (Ch 89), or over-fetching.

93.1 Problem Definition and Real-World Context

Lists are the **most common** and most **scroll-performance-sensitive** UI: a product catalog, a feed (Ch 71), order history, search results (Ch 9) — often **thousands** of items. Rendering them **all** is impossible (memory, Ch 89) and laying out off-screen items wastes work. **Large list performance** is the discipline of scrolling big lists **smoothly** (every frame within budget, Ch 90) by **only composing/laying out visible items** (lazy lists recycle/reuse), giving items **stable** `key`**s** and `contentType` (correct, efficient reuse), keeping each item composable **light and stable** (Ch 90 — no over-recomposition, no heavy work), loading **images asynchronously** (Coil, Ch 59 — never block the scroll on decode), and **paging** data (Ch 71 — don't load everything). The classic failures are **jank on scroll** (heavy/unstable items, sync image decode), **memory bloat** (not recycling, full-res bitmaps, Ch 89/59), and **over-fetching** (loading all data instead of pages, Ch 71).

Concretely, teams hit:

- **Rendering all items (memory + layout death).** Putting thousands of items in a non-lazy container (`Column` with all items, or a `ScrollView`) composes/lays out **everything** → huge memory (Ch 89) and slow layout. **Fix: lazy lists** (`LazyColumn` / `LazyRow` / `LazyVerticalGrid`) that only compose **visible** items (+ a small buffer) and **recycle** as you scroll.
- **Missing / unstable** `key`**s (the #1 list bug).** Without a **stable** `key` per item, Compose can't track item **identity** across data changes — so inserts/removes/moves cause **wrong reuse** (state attached to the wrong item), **unnecessary recomposition**, and **scroll-position jumps**. **Fix:** `items(list, key = { it.id })` with a **stable, unique** id.
- **Heavy / unstable item composables (Ch 90).** An item that does **expensive work** (formatting, allocation, sorting) **in composition**, or that **over-recomposes** (unstable params, Ch 90), janks on **every** scroll frame as items compose. **Fix:** light, **skippable** items (stable params, hoisted work, Ch 90).
- **Synchronous image decode on scroll (Ch 59).** Decoding a bitmap **synchronously** in the item (or on the main thread) **blocks the frame** → scroll stutter. **Fix: async** image loading (**Coil** `AsyncImage`, Ch 59) that decodes off-main, downsamples to view size (Ch 89), caches, and cancels on recycle.

- **No `contentType` for mixed lists.** A list with **different item types** (header, product, ad) reuses incompatible layouts inefficiently without `contentType`. **Fix:** provide `contentType` so Compose reuses the **right** item structure.
- **Over-fetching / loading all data (Ch 71).** Loading the **entire** dataset into memory to show a list wastes memory (Ch 89) and network (Ch 91). **Fix: Paging 3** (Ch 71) — load **pages** as the user scrolls, with placeholders and prefetch distance.
- **Expensive scroll work.** Recomputing derived values, triggering side effects, or doing I/O **per scroll** janks. **Fix:** `derivedStateOf` for scroll-driven state (Ch 90), hoist work, no I/O in composition (Ch 90/3).
- **First-scroll JIT jank (Ch 92).** The first scroll runs JIT-cold → janky. **Fix:** a **Baseline Profile** covering the scroll path (Ch 92).

The architecture: scroll large lists smoothly with **lazy lists** (`LazyColumn` / `LazyRow` / `LazyVerticalGrid` — compose only visible items, recycle) + **stable `key`s** (correct identity/ reuse, no position jumps) + **`contentType`** (efficient mixed-type reuse) + **light, stable item composables** (skippable, hoisted work, Ch 90) + **async image loading** (Coil off-main + downsample + cache + cancel, Ch 59/89) + **paging** (Paging 3, load pages not everything, Ch 71) + **scroll-state discipline** (`derivedStateOf`, no per-scroll work, Ch 90) + a **Baseline Profile** for the scroll path (Ch 92) — verified with **JankStats/Macrobenchmark `FrameTimingMetric`** (Ch 85/90). This chapter is about **smooth scrolling of big lists**.

Three practical use cases

1. **Smooth product catalog grid (StreamCart).** 5,000 products in a `LazyVerticalGrid` with `key = { it.id }`, `contentType`, light `@Immutable` item cards (Ch 90), Coil `AsyncImage` (Ch 59), and Paging 3 (Ch 71) — scrolls within budget, bounded memory (Ch 89).
2. **Fix a janky feed (Ch 71).** The feed janked from heavy item composables + sync image decode; fix: lighten items (hoist work, stable params, Ch 90) + Coil async images (Ch 59) + stable keys → smooth.
3. **Paged order history (Ch 71).** Thousands of orders loaded via Paging 3 with placeholders and prefetch — no over-fetch, no memory blowup (Ch 89/91).

Why this is architectural

Large list performance composes recomposition/jank (Ch 90), paging/infinite feed (Ch 71), image loading (Ch 59), memory (Ch 89), baseline profiles (Ch 92), and measurement (Ch 85). The decisions — lazy recycling, stable keys + contentType, light stable items, async images, paging — determine whether the most common UI scrolls smoothly or janks and bloats.

93.2 Requirements Engineering

Functional requirements

ID	Requirement
F1	Use lazy lists (compose only visible items, recycle) — never render all items.
F2	Stable `key`s per item (correct identity/reuse; no position jumps).
F3	`contentType` for mixed-type lists (efficient reuse).
F4	Light, stable, skippable item composables; hoist work (Ch 90).
F5	Async image loading (Coil: off-main, downsample, cache, cancel) (Ch 59/89).
F6	Page data (Paging 3) instead of loading everything (Ch 71); verify with FrameTiming/JankStats (Ch 85/90).

Non-functional requirements

Attribute	Policy
Smoothness	Every scroll frame within budget; low jank (Ch 90/85).
Memory	Bounded (recycling + downsampled images + paging, Ch 89).
Efficiency	Paged fetch; no over-fetch (Ch 71/91).
Correctness	Stable keys → correct item state across changes.
Measured	FrameTiming (lab) + JankStats (field) (Ch 85/90).

Scope

- **In scope:** lazy lists (`LazyColumn` / `LazyRow` / `LazyVerticalGrid`) + recycling, stable `key`s (identity/reuse/position), `contentType`, light + stable + skippable items (Ch 90), async images (Coil, Ch 59) + downsampling (Ch 89), paging (Paging 3, Ch 71), scroll-state discipline (`derivedStateOf` , Ch 90), prefetch distance, baseline profile for scroll (Ch 92), measurement (FrameTiming/JankStats, Ch 85/90).
- **Out of scope:** jank/recomposition detail (Ch 90), paging detail (Ch 71), image-loader internals (Ch 59), memory detail (Ch 89), baseline-profile mechanics (Ch 92), Macrobenchmark setup (Ch 85).

Assumptions & constraints

- Jetpack Compose lazy lists (Ch 2/90); Coil for images (Ch 59); Paging 3 for data (Ch 71); items have **stable unique ids**. Measured with Macrobenchmark `FrameTimingMetric` (lab) + JankStats (field, Ch 85/90). Baseline Profile covers scroll (Ch 92). `minSdk 24` .

93.3 System Design Fundamentals

- **Lazy lists & recycling (the foundation).** `LazyColumn` / `LazyRow` / `LazyVerticalGrid` (Compose) — like `RecyclerView` (Views) — **only compose and lay out the items currently visible** (plus a small buffer), and

reuse/recycle item slots as you scroll. This keeps **memory bounded** (Ch 89) and **layout cheap** regardless of dataset size. **Never** put a large dataset in a non-lazy `Column`/`ScrollView` (composes/lays out **everything**). Lazy is the baseline for any list that could be long.

- **Stable `key`s — identity, reuse, and position (the #1 lever).** By default a lazy list keys items **by position/index**; when the data changes (insert/remove/move), position-keying causes **wrong reuse** — item state (scroll position within item, animations, remembered values) sticks to the **wrong** item, items **recompose** unnecessarily, and the **scroll position jumps**. Providing a **stable, unique** `key` (`items(list, key = { it.id })`) lets Compose track each item's **identity** → correct reuse, minimal recomposition, stable scroll position across data changes, and correct item animations. **Always provide stable keys** for dynamic lists.
- **`contentType` — efficient reuse of mixed items.** When a list has **different item types** (header vs product vs ad vs footer), Compose reuses item compositions more efficiently if you declare `contentType` per item — so it reuses a **product** slot for a product, not for a header. For single-type lists it's optional; for **mixed** lists it **improves reuse** and reduces recomposition. Provide `contentType` for heterogeneous lists.
- **Light, stable, skippable item composables (Ch 90).** Each item composes as it scrolls into view, so item cost is paid **repeatedly**. An item that **over-recomposes** (unstable params, Ch 90) or does **heavy work in composition** (formatting, allocation, sorting, I/O) **janks** the scroll. Make items **skippable** (stable/`@Immutable` params, Ch 90), **hoist** work out of the item (precompute in the `ViewModel`/paging map, Ch 90/3), and keep item layout **shallow** (deep nesting costs layout). The item composable is the hot path — keep it **light and stable**.
- **Async image loading (Ch 59/89).** Images are the **most common scroll-jank source** and the **biggest memory risk** (Ch 89). **Decode off the main thread** (never synchronously in the item), **downsample** to the view size (Ch 89 — avoid full-res bitmaps → OOM), **cache** (memory/disk, Ch 59), and **cancel** the load when the item is recycled/scrolled away. **Coil**'s `AsyncImage` does all of this — use it; **never** decode bitmaps synchronously in a list item.
- **Paging — don't load everything (Ch 71).** Loading the **entire** dataset to show a list wastes **memory** (Ch 89) and **network/battery** (Ch 91). **Paging 3** (Ch 71) loads **pages** as the user scrolls (with a **prefetch distance** so the next page is ready before they hit the end), supports **placeholders**, and integrates with lazy lists (`collectAsLazyPagingItems`). For any potentially large/remote dataset, **page it** (Ch 71).
- **Scroll-state discipline (Ch 90).** Reading scroll state (`LazyListState.firstVisibleItemIndex`/`ScrollOffset`) **directly** in composition recomposes **every scroll frame**; wrap scroll-driven derived state in `derivedStateOf` (recompute only when the *derived* value changes, Ch 90), and **defer** visual effects (parallax) to `graphicsLayer` lambdas (read in draw, not composition, Ch 90). **Never** do I/O or heavy work per scroll.
- **Prefetch & smoothness tuning.** Lazy lists **prefetch** upcoming items; Paging's **prefetch distance** loads the next page ahead of time (Ch 71). Keep item composition cheap enough that prefetch stays ahead of the scroll. Avoid **unbounded** item heights that thrash layout.
- **First-scroll JIT jank → Baseline Profile (Ch 92).** The **first** scroll runs **JIT-cold** (interpreted/JIT Compose + scroll code) → janky until warm. A **Baseline Profile** covering the **scroll journey** (Ch 92) AOT-compiles it → **smooth from the first scroll**. Include lists in the profile.
- **Measurement (Ch 85/90).** Verify scroll smoothness with **Macrobenchmark** `FrameTimingMetric` (lab — fling the list, measure frame durations, Ch 90) and **JankStats** (field — per-screen jank, Ch 85). Use **Layout Inspector** to find over-recomposing items (Ch 90). **Measure** the scroll — don't assume.

Core insight: **scroll large lists smoothly by never rendering everything — use lazy lists (LazyColumn/LazyRow/LazyVerticalGrid, like RecyclerView) that compose and lay out only the visible items and recycle slots as you scroll, keeping memory bounded (Ch 89) and layout cheap regardless of dataset size;**

give every item a stable, unique key (items(list, key = { it.id })) so Compose tracks identity across inserts/removes/moves — otherwise position-keying causes wrong reuse (state on the wrong item), needless recomposition, and scroll-position jumps — and declare contentType for mixed-type lists so the right slot is reused; keep each item composable light, stable, and skippable (@Immutable params, hoisted work, shallow layout, Ch 90) because item cost is paid repeatedly as items scroll into view, and load images asynchronously with Coil (off-main decode, downsample to view size, cache, cancel on recycle, Ch 59/89) because synchronous bitmap decode is the top scroll-jank and OOM source; page the data with Paging 3 (load pages with a prefetch distance, not the whole dataset, Ch 71) to bound memory and network (Ch 89/91); keep scroll-driven state cheap with derivedStateOf and graphicsLayer deferral and never do I/O per scroll (Ch 90); ship a Baseline Profile covering the scroll path so the first scroll isn't JIT-janky (Ch 92); and verify with Macrobenchmark FrameTimingMetric (lab) and JankStats (field, Ch 85/90) — because lists are the most common and most scroll-sensitive UI, and jank, memory bloat, and over-fetching all show up here.

93.4 High-Level Architecture


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Components & ownership

- **Paging 3** loads pages + prefetch (not everything, Ch 71); **lazy list** composes only visible
    - recycles.
- **Stable keys** (identity/reuse) + `contentType` (mixed reuse); **light stable items** (Ch 90).
- **Async images** (Coil off-main/downsample/cache/cancel, Ch 59/89); **scroll discipline** (`derivedStateOf` , Ch 90).
- **Baseline Profile** for scroll (Ch 92); **verified** (FrameTiming/JankStats, Ch 85/90).

Direction: page → lazy + recycle → stable keys + contentType → light stable items → async images + scroll discipline → smooth; profiled + measured.

93.5 Detailed Component Design


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Component	Responsibility
`LazyListSetup`	lazy list + stable keys + `contentType` (visible-only + recycle)
`ItemComposable`	light, stable, skippable item; hoisted work (Ch 90)
`AsyncImageLoad`	Coil off-main decode + downsample + cache + cancel (Ch 59/89)
`PagedData`	Paging 3 pages + prefetch + placeholders (Ch 71)
`ScrollState`	`derivedStateOf` + `graphicsLayer` ; no per-scroll work (Ch 90)

93.6 End-to-End Data Flow

Scroll → recycle → compose visible → async image



Stable key vs position key (on data change)



93.7 Complete Kotlin Implementation

Package: `com.streamcart.ui.list`.

93.7.1 Lazy list + stable keys + contentType

```
@Composable
fun ProductCatalog(items: LazyPagingItems<CatalogRow>) {   // Paging 3 (Ch 71)
    LazyVerticalGrid(columns = GridCells.Adaptive(160.dp)) {
        items(
            count = items.itemCount,
            key = items.itemKey { it.id },                  // STABLE key -> identity/reuse, no
jumps
            contentType = items.itemContentType { it.type }, // mixed types (product/header/ad)
-> efficient reuse
        ) { index ->
            when (val row = items[index]) {
                is CatalogRow.Product -> ProductCard(row.product)   // light, stable item (Ch
90)
                is CatalogRow.Header -> SectionHeader(row.title)
                null -> PlaceholderCard()                           // Paging placeholder (Ch
71)
            }
        }
    }
}
```

93.7.2 Light, stable, skippable item + async image (Ch 90/59)

```
@Immutable                                                  // stable -> skippable (Ch 90)
data class ProductUi(val id: String, val title: String, val price: String, val imageUrl: String)

@Composable
fun ProductCard(product: ProductUi) {
    // Skippable item (stable param). Shallow layout. NO heavy work / NO I/O in composition (Ch
90).
    Card {
        AsyncImage(                                         // Coil: off-main decode,
downsample, cache, cancel (Ch 59/89)
            model = ImageRequest.Builder(LocalContext.current)
                .data(product.imageUrl).size(Size.ORIGINAL).build(),
            contentDescription = product.title,
            modifier = Modifier.aspectRatio(1f),            // bounded size -> downsample target
(Ch 89)
            contentScale = ContentScale.Crop,
        )
        Text(product.title); Text(product.price)            // price pre-formatted in the
ViewModel/paging map (Ch 90)
    }
}
// Pre-format (price, dates) in the ViewModel/PagingData.map - NOT in the item composable (Ch
90).
```

93.7.3 Scroll-state discipline (derivedStateOf, Ch 90)

```
@Composable
fun CatalogWithScrollToTop(listState: LazyGridState, items: LazyPagingItems<CatalogRow>) {
    // derivedStateOf: FAB recomposes only when crossing the threshold, NOT every scroll pixel
(Ch 90)
    val showScrollTop by remember { derivedStateOf { listState.firstVisibleItemIndex > 6 } }
    Box {
        LazyVerticalGrid(state = listState, columns = GridCells.Adaptive(160.dp)) { /* items as
above */ }
        if (showScrollTop) ScrollToTopFab(listState)
    }
    // NEVER read firstVisibleItemIndex directly in composition (recomposes every frame), and NO
I/O per scroll.
}
```

93.7.4 Paging source (don't load everything, Ch 71)

```
class CatalogViewModel @Inject constructor(repo: CatalogRepository) : ViewModel() {
    // Paging 3: pages loaded on scroll with prefetch; not the whole dataset (Ch 71)
    val catalog: Flow<PagingData<CatalogRow>> = Pager(
        config = PagingConfig(pageSize = 30, prefetchDistance = 10, enablePlaceholders = true),
// prefetch ahead
    ) { repo.catalogPagingSource() }
        .flow.map { it.map { product -> product.toUi() } }   // pre-format off the item (Ch
90/3)
        .cachedIn(viewModelScope)
}
```

93.7.5 Verify scroll smoothness (Macrobenchmark FrameTiming, Ch 85/90)

```
@RunWith(AndroidJUnit4::class)
class CatalogScrollBenchmark {
    @get:Rule val rule = MacrobenchmarkRule()
    @Test fun scroll() = rule.measureRepeated(
        packageName = "com.streamcart", metrics = listOf(FrameTimingMetric()),   // frame
durations (Ch 90)
        compilationMode = CompilationMode.Partial(), iterations = 10, startupMode = null,  //
baseline profile (Ch 92)
    ) {
        startActivityAndWait()
        val grid = device.findObject(By.res("catalog"))
        repeat(5) { grid.fling(Direction.DOWN) }                                  // assert frame
p95 within budget
    }
    // Field: JankStats per screen (Ch 85). Dev: Layout Inspector for over-recomposing items (Ch
90).
}
```

93.7.6 Module structure

```
:feature:catalog  ProductCatalog (lazy + keys + contentType), ProductCard (light/stable, Ch 90),
CatalogViewModel (Paging 3, Ch 71)
# Recycling -> lazy lists; images -> Coil (Ch 59/89); paging -> Ch 71; item stability -> Ch 90;
# scroll path AOT -> Baseline Profile (Ch 92); verify -> FrameTiming/JankStats (Ch 85/90).
```

93.8 Concurrency and Race Conditions

Scenario	Risk	Handling
Rendering all items	memory + layout death	lazy lists (visible-only + recycle)
Position keys on data change	wrong reuse + scroll jump	stable `key = { it.id }`
Sync image decode on scroll	frame block	Coil async off-main (Ch 59)
Heavy/unstable item	per-frame jank	light + skippable item (Ch 90)
Full-res bitmaps	OOM	downsample to view size (Ch 89)
Load-everything	memory/network bloat	Paging 3 (Ch 71)
Scroll-state read in composition	recompose every frame	derivedStateOf / graphicsLayer (Ch 90)
Image load after recycle	wasted/wrong image	cancel on recycle (Coil, Ch 59)

- **Stable keys fix a whole class of correctness + perf bugs at once:** identity-based reuse means item state stays with the right item, recomposition is minimal, scroll position is stable across data changes, and item animations are correct — missing keys break all four.
- **Async image loading is the single biggest scroll-smoothness lever (Ch 59/89):** sync decode blocks the frame and full-res bitmaps OOM; Coil's off-main decode + downsample + cache + cancel- on-recycle solves both, and cancellation prevents wasted work on fast scroll.
- **Paging bounds memory and network together (Ch 71/89/91):** loading pages (not everything) keeps the in-memory list bounded and avoids over-fetching — essential for large/remote datasets.

93.9 Failure Scenarios and Recovery

Failure	Detection	Response	Outcome
Scroll jank	JankStats/FrameTiming (Ch 85/90)	light items + async images + stable keys (Ch 90/59)	smooth
Scroll-position jumps	visual/state bugs	stable `key = { it.id }`	stable
OOM on long list	field OOM (Ch 89/83)	downsample + recycle + paging (Ch 89/71)	bounded
Over-fetch / slow load	network/memory (Ch 91)	Paging 3 (Ch 71)	paged
Item over-recomposes	Layout Inspector (Ch 90)	stable params + hoist work (Ch 90)	skippable
First-scroll jank	Macrobenchmark None vs Partial	Baseline Profile scroll path (Ch 92)	AOT smooth
Wrong image on fast scroll	visual	cancel on recycle (Coil, Ch 59)	correct


93.9 Failure Scenarios and Recovery

> **Lazy + stable keys + light items + async images + paging.** Large-list jank comes from rendering too much (lazy + paging fix it), position-keyed reuse (stable keys fix it), heavy/ unstable items (light skippable items fix it, Ch 90), and sync image decode (Coil async fixes it, Ch 59). Bound memory with recycling + downsampling + paging (Ch 89/71), AOT the scroll path (Ch 92), and verify with FrameTiming/JankStats (Ch 85/90).

93.10 Scaling the Implementation

- **Level 1.** Lazy list (`LazyColumn` /grid) with **stable keys**; Coil `AsyncImage` (Ch 59).
- **Level 2.** `contentType` for mixed lists; light `@Immutable` items (Ch 90); `derivedStateOf` for scroll state; Paging 3 for large/remote data (Ch 71).
- **Level 3 — production.** Baseline Profile covering scroll (Ch 92); JankStats field monitoring (Ch 85); Macrobenchmark `FrameTimingMetric` CI gate (Ch 90); downsampled images (Ch 89).
- **Level 4 — large.** Prefetch tuning, placeholders/skeletons, item-type optimization, nested- scroll performance, sticky headers, per-screen jank budgets (Ch 85).
- **Level 5 — enterprise.** List-perf SLOs, automated frame-regression gates (Ch 92), device-tiered item complexity, continuous field jank dashboards by list screen (Ch 85).

Avoid overengineering: lazy list + stable keys + light items + Coil async images + Paging 3, verified with FrameTiming/JankStats, covers virtually every list. Don't render all items; don't position-key; don't decode images synchronously; don't load everything; don't skip measurement.

93.11 Alternative Approaches and Trade-offs

Dimension	Option A	Option B	When
Container	Lazy list (recycle)	non-lazy Column/ScrollView	Lazy for any long list
Keys	Stable `key = id`	position/index	Stable (correct reuse + scroll)
Mixed items	`contentType`	none	contentType for heterogeneous
Item	Light/stable/skippable (Ch 90)	heavy/unstable	Light (per-frame cost)
Images	Coil async (Ch 59)	sync decode	Async (no frame block)
Data	Paging 3 (Ch 71)	load all	Page large/remote datasets
Scroll state	derivedStateOf (Ch 90)	direct read	derivedStateOf (no recompose storm)

Lazy vs non-lazy container: a non-lazy `Column` / `ScrollView` composes and lays out **every** item → memory + layout blow up on large lists. **Lazy lists** compose only **visible** items and **recycle** → bounded memory, cheap layout, any size. **Always lazy** for lists that could be long.

Stable keys vs position keys: position-keying breaks on data change — **wrong reuse**, **state on the wrong item**, **scroll jumps**, **extra recomposition**. **Stable unique keys** give **identity-based** reuse → correct, minimal, stable. Always provide stable keys for dynamic lists.

Async images vs sync decode (Ch 59): synchronous bitmap decode in an item **blocks the scroll frame** and risks **OOM** (full-res, Ch 89). **Coil** decodes **off-main**, **downsamples**, **caches**, and **cancels on recycle** → smooth + bounded. Never decode images synchronously in a list.

Paging vs load-all (Ch 71): loading the whole dataset wastes **memory** (Ch 89) and **network/ battery** (Ch 91). **Paging 3** loads pages with prefetch → bounded + efficient. Page any large or remote dataset.

93.12 Security & Privacy Considerations

- **Image URLs / content in lists (Ch 59/28):** list items often show user/remote content — load images through the **managed loader** (Coil, Ch 59) with proper caching/headers; don't leak auth in image URLs; respect content visibility (Ch 28).
- **Paged data respects authorization (Ch 71/20):** each page fetch is an **authorized** request (Ch 20); don't cache another user's paged data across accounts (per-account isolation, Ch 28).
- **Placeholders don't leak stale data:** ensure placeholders/recycled slots don't briefly show a **previous item's** sensitive content (clear on reuse) — stable keys + correct reuse help here.
- **No PII in scroll/jank telemetry (Ch 85/28):** JankStats/FrameTiming report **timing + screen**, not item content (Ch 28).
- **Sensitive lists (Ch 76):** a list showing sensitive data (payment methods, Ch 76) renders under the same **secure-surface** rules — orthogonal to list perf.
- Client-mostly concern: list performance is **client-side**; the cross-cutting notes are **authorized paging** (Ch 20/71), **per-account isolation** (Ch 28), and **no stale content on recycle**.

93.13 Performance and Resource Optimization

- **This chapter *is* list performance** — summary: **lazy** (visible-only + recycle), **stable keys**, `contentType`, **light/stable/skippable items** (Ch 90), **async images** (Coil off- main + downsample + cache + cancel, Ch 59/89), **paging** (Ch 71), `derivedStateOf` scroll state (Ch 90), **Baseline Profile** scroll path (Ch 92).
- **Keep item composition cheap** — hoist formatting/work into the `ViewModel`/paging map (Ch 90); shallow layout.
- **Downsample images to slot size** — biggest memory lever (Ch 89).
- **Page + prefetch** — bounded memory/network; next page ready before the end (Ch 71).
- **Verify the scroll** — FrameTiming (lab, Ch 90) + JankStats (field, Ch 85); Layout Inspector for over-recomposing items (Ch 90).
- **Profiling:** frame p95 on scroll (FrameTiming/JankStats), item recomposition counts (Layout Inspector), list memory footprint (Ch 89), image cache hit rate (Ch 59), page-load timing (Ch 71).

93.14 Testing Strategy

Scroll smoothness is measured (Macrobenchmark); keys/paging/item stability are verified.

```
class CatalogScrollBenchmark {   // lab gate (Ch 90/92)
    @get:Rule val rule = MacrobenchmarkRule()
    @Test fun scroll_frames() = rule.measureRepeated(
        packageName = "com.streamcart", metrics = listOf(FrameTimingMetric()),
        compilationMode = CompilationMode.Partial(), iterations = 10, startupMode = null,
    ) { startActivityAndWait(); repeat(5) {
device.findObject(By.res("catalog")).fling(Direction.DOWN) } }
    // assert frameDurationCpuMs p95 within budget
}

class PagingTest {
    @Test fun `loads pages not everything`() = runTest { /* only pageSize+prefetch loaded
initially (Ch 71) */ }
}
@Test fun `stable keys keep scroll position on insert`() { /* insert item -> position stable, no
jump */ }
// Dev: Layout Inspector -> item recomposition ~1 per data change (Ch 90); compiler stability
(item @Immutable).
// Field: JankStats per list screen (Ch 85).
```

What to test: Macrobenchmark `FrameTimingMetric` on fling (lab gate, p95 within budget, Ch 90/92); Paging loads pages not everything (Ch 71); stable keys keep scroll position across inserts; item recomposition minimal (Layout Inspector, Ch 90); item `@Immutable`/skippable (compiler stability, Ch 90); async image (no sync decode, Ch 59); field jank via JankStats (Ch 85).

93.15 Production Observability

- **Metrics: scroll jank rate** + **frozen frames** per list screen (JankStats, field, Ch 85), **frame p95 on scroll** (FrameTiming, lab, Ch 90), list **memory footprint** (Ch 89), **image cache hit rate** + decode time (Ch 59), **page-load latency** + prefetch effectiveness (Ch 71), first-scroll jank (baseline-profile effect, Ch 92).
- **Logs/traces:** scroll frame traces (Perfetto, dev); paging load events (Ch 71) — PII-safe (Ch 28).
- **Debugging:** scroll jank → JankStats (which list, Ch 85) → Layout Inspector (over-recomposing item? Ch 90) → is image sync-decoded? (Ch 59) → heavy item work? (Ch 90); position jumps → stable keys?; OOM → downsample/recycle/paging (Ch 89/71); over-fetch → Paging config (Ch 71); first- scroll jank → `None` vs `Partial` (Ch 92).
- **Alerting:** list-screen jank/frozen regression release-over-release (Ch 85/41); frame p95 budget breach (Macrobenchmark CI, Ch 90/92); list OOM (Ch 89/83).
- **Dashboards:** jank/frozen by list screen/device/release; scroll frame percentiles; list memory; image cache; page-load latency.

93.16 System Design Interview Preparation

> ≥25 questions.

Fundamentals

1. *Why not render all items?* Memory + layout blow up; use lazy lists.
2. *Lazy list?* Composes only visible items + recycles (like RecyclerView).
3. *Compose lazy APIs?* `LazyColumn` / `LazyRow` / `LazyVerticalGrid`.
4. *Biggest list bug?* Missing/unstable keys.
5. *Why keys?* Identity → correct reuse, no scroll jumps, minimal recomposition.

Keys/contentType

6. *Default key?* Position/index — breaks on data change.
7. *Stable key?* `items(list, key = { it.id })` — unique + stable.
8. *What breaks without keys?* Wrong reuse, state on wrong item, scroll jump, extra recomposition.
9. *contentType?* Efficient reuse for mixed item types.
10. *Single-type list?* contentType optional.

Items

11. *Item cost?* Paid repeatedly as items scroll in — keep light.
12. *Over-recomposing item?* Stable/@Immutable params (Ch 90).
13. *Heavy work in item?* Hoist to ViewModel/paging map (Ch 90).
14. *Shallow layout?* Deep nesting costs layout.

Images

15. *Image jank?* Sync decode blocks the frame.
16. *Fix?* Coil async — off-main decode (Ch 59).
17. *Memory?* Downsample to view size (Ch 89).
18. *Fast scroll?* Cancel load on recycle (Ch 59).

Paging

19. *Load everything?* No — Paging 3 (Ch 71).
20. *Prefetch distance?* Load next page before the end.
21. *Placeholders?* Show skeletons for not-yet-loaded (Ch 71).
22. *Why page?* Bound memory + network (Ch 89/91).

Scroll/measure

23. *Scroll-state read?* `derivedStateOf` — not direct (Ch 90).
24. *First-scroll jank?* Baseline Profile scroll path (Ch 92).
25. *Measure?* Macrobenchmark `FrameTimingMetric` (lab) + JankStats (field) (Ch 85/90).
26. *Find slow item?* Layout Inspector recomposition counts (Ch 90).

Design-change challenges

1. **Smooth a 5,000-item grid.** Lazy grid + stable keys + contentType + light items + Coil + Paging 3 (Ch 90/59/71).
2. **Fix scroll-position jumps.** Stable `key = { it.id }`.
3. **Stop OOM on a long image list.** Downsample + recycle + paging (Ch 89/71).
4. **Smooth the first scroll.** Baseline Profile scroll path (Ch 92).
5. **Gate scroll-jank regressions.** Macrobenchmark `FrameTimingMetric` (Ch 90) + field JankStats (Ch 85).

Mock walkthrough (abridged)

Clarify (dataset size, remote?, mixed types?, device targets?) → lazy list (visible-only + recycle) → stable keys + contentType → light/stable/skippable items (Ch 90) → async images (Coil off-main + downsample + cancel, Ch 59/89) → Paging 3 (pages + prefetch + placeholders, Ch 71) → scroll discipline (derivedStateOf, no per-scroll work, Ch 90) → Baseline Profile scroll path (Ch 92) → verify FrameTiming (lab) + JankStats (field) (Ch 85/90) → observability (jank by list screen).

93.17 Practical Exercises

1. **Beginner.** Build a `LazyColumn` with stable keys (`key = { it.id }`) and Coil `AsyncImage` (Ch 59); confirm scroll position is stable on insert. *Eval:* keyed; async images; no jump.
2. **Intermediate.** Add `contentType` for a mixed list (headers + products), make items `@Immutable`/skippable, and hoist price formatting out of the item (Ch 90). *Eval:* efficient reuse; minimal item recomposition.
3. **Advanced.** Back the list with Paging 3 (prefetch + placeholders, Ch 71) and add a Baseline Profile covering the scroll (Ch 92); verify with Macrobenchmark `FrameTimingMetric`. *Eval:* paged; smooth first scroll.
4. **Production debugging.** Use JankStats (Ch 85) to find a janky list, then Layout Inspector to find an over-recomposing item (Ch 90) and a sync-decoded image (Ch 59); fix both. *Eval:* root causes found and fixed.
5. **System design challenge.** Design a list-perf budget + CI frame-gating (Macrobenchmark per list screen) + field JankStats dashboards by device class (Ch 85/92). *Eval:* gated, measured, device-tiered.

(Solutions withheld unless requested.)

93.18 Chapter Summary & Reference Sheet

Key decisions: scroll large lists smoothly by **never rendering everything** — use **lazy lists** (**`LazyColumn`** / **`LazyRow`** / **`LazyVerticalGrid`**, like `RecyclerView`) that **compose and lay out only the visible items and recycle** slots as you scroll, keeping **memory bounded** (Ch 89) and layout cheap regardless of dataset size; give every item a **stable, unique `key`** (`items(list, key = { it.id })`) so Compose tracks **identity** across inserts/removes/moves — otherwise position-keying causes **wrong reuse** (state on the wrong item), needless recomposition, and **scroll-position jumps** — and declare **`contentType`** for **mixed-type** lists so the right slot is reused; keep each **item composable light, stable, and skippable** (**`@Immutable`** params, **hoisted** work, shallow layout, Ch 90) because item cost is paid **repeatedly** as items scroll into view; load images **asynchronously** with **Coil** (off-main decode, **downsample** to view size, cache, **cancel on recycle**, Ch 59/89) because **synchronous bitmap decode** is the top scroll-jank and OOM source; **page** the data with **Paging 3** (load pages with a **prefetch**

distance, not the whole dataset, Ch 71) to bound memory and network (Ch 89/91); keep scroll-driven state cheap with `derivedStateOf` and `graphicsLayer` deferral and **never do I/O per scroll** (Ch 90); ship a **Baseline Profile** covering the **scroll path** so the first scroll isn't **JIT-janky** (Ch 92); and **verify** with **Macrobenchmark** `FrameTimingMetric` (lab) and **JankStats** (field, Ch 85/90) — because lists are the **most common and most scroll-sensitive** UI, and **jank, memory bloat, and over-fetching** all show up here.

Concepts: lazy lists + recycling (`LazyColumn` / `LazyRow` / `LazyVerticalGrid`), stable `key`s (identity/reuse/position), `contentType` (mixed reuse), light/stable/skippable items (Ch 90), async images (Coil off-main/downsample/cache/cancel, Ch 59/89), Paging 3 (pages + prefetch + placeholders, Ch 71), scroll-state discipline (`derivedStateOf` / `graphicsLayer` , Ch 90), Baseline Profile scroll path (Ch 92), measurement (FrameTiming/JankStats, Ch 85/90).

Common mistakes: rendering all items (non-lazy container); missing/unstable keys (wrong reuse + scroll jumps); no `contentType` on mixed lists; heavy/unstable item composables (Ch 90); synchronous image decode (jank + OOM, Ch 59/89); loading everything instead of paging (Ch 71); reading scroll state directly in composition (recompose storm, Ch 90); I/O per scroll; no Baseline Profile (first- scroll JIT jank, Ch 92); not measuring the scroll.

Checklist:

- [ ] lazy list (visible-only + recycle)
- [ ] stable `key = { it.id }` (no position jumps)
- [ ] `contentType` for mixed lists
- [ ] light + stable + skippable items (`@Immutable` , hoisted work, Ch 90)
- [ ] Coil async images (off-main + downsample + cache + cancel, Ch 59/89)
- [ ] Paging 3 (pages + prefetch + placeholders, Ch 71)
- [ ] `derivedStateOf` / `graphicsLayer` scroll state; no per-scroll I/O (Ch 90)
- [ ] Baseline Profile covers scroll (Ch 92)
- [ ] Macrobenchmark `FrameTimingMetric` gate (Ch 90) + JankStats field (Ch 85)
- [ ] clear sensitive content on recycle (Ch 28).

Key APIs: `LazyColumn` / `LazyRow` / `LazyVerticalGrid` (+ `items(..., key, contentType)`), `LazyPagingItems` (`itemKey` / `itemContentType` / `collectAsLazyPagingItems` , Ch 71), `Pager` + `PagingConfig` (`pageSize` / `prefetchDistance` / `enablePlaceholders` , Ch 71), `cachedIn` , Coil `AsyncImage` / `ImageRequest` (Ch 59), `@Immutable` items (Ch 90), `derivedStateOf` + `LazyListState` / `LazyGridState` (Ch 90), `Modifier.graphicsLayer` (defer), `MacrobenchmarkRule` + `FrameTimingMetric` (Ch 90/92), JankStats (Ch 85), Layout Inspector.

References:

- Lazy lists: https://developer.android.com/develop/ui/compose/lists
- Compose list performance / keys: https://developer.android.com/develop/ui/compose/lists#item-keys
- Paging 3: https://developer.android.com/topic/libraries/architecture/paging/v3-overview
- Coil (async images): https://coil-kt.github.io/coil/compose/
- (composes) Ch 90 Jank/Recomposition, Ch 71 Infinite Feed/Paging, Ch 59 Image Loading, Ch 89 Memory, Ch 85 Perf, Ch 92 Baseline Profiles, Ch 2 Compose

Chapter review (Phase 4)

- **Completeness:** all 18 sections; architecture/class/sequence/key-vs-position/failure diagrams; lazy+keys+contentType + light-item+async-image + scroll-discipline + paging + Macrobenchmark code; concurrency (stable-keys-fix-class, async-images-lever, paging-bounds); failure (lazy+keys+light+async+paging principle); 5 levels; lazy-vs-nonlazy, keys, async-images, paging trade-offs; security (authorized paging, per-account isolation, no stale on recycle); perf (the whole chapter); FrameTiming + paging + keys tests; observability (jank by list screen); 26 Q&A + 5 challenges. ✅
- **Expand later:** jank/recomposition (Ch 90), paging (Ch 71), image loading (Ch 59), memory (Ch 89), baseline profiles (Ch 92).
- **Interview coverage:** exceeds ≥25 bar.